

Max Callaghan

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Education

- 2018–2022 **PhD**, *University of Leeds*, School of Earth and Environment
Machine Reading the Science of Climate Change: Computational Tools to Support Evidence-Based Decision-Making in the Age of Big Literature
Supervisors: Professor Jan Minx, Professor Piers Forster
- 2013–2016 **Master of Public Policy**, *Hertie School of Governance*, Berlin, 91/100
- 2007–2011 **BA - English and Related Literature**, *University of York*, First-class honours

Employment History

- 2016– **Researcher**, *Mercator Research on Global Commons and Climate Change (MCC)*, Berlin
- 2015–2016 **Research Assistant**, *Mercator Research on Global Commons and Climate Change (MCC)*, Berlin
- 2016–2016 **Teaching Assistant – Stats II**, *Hertie School of Governance*, Berlin
- 2015–2015 **Teaching Assistant – Applied Economic Analysis**, *Hertie School of Governance*
- 2014–2015 **Junior Research Fellow – Development, Gender and Human Rights**, *United Nations System Staff College (UNSSC)*, Turin
- 2014–2014 **Research Assistant – The Governance Report**, *Hertie School of Governance*, Berlin
- 2013–2014 **Research Analyst**, *Market Logic Software*, Berlin
- 2013–2013 **Junior Consultant**, *Cadence Partnership*, London
- 2012–2013 **Communications Intern**, *Release*, London
- 2011–2012 **Teacher**, *The English School of Mongolia*, Ulaanbaator

Honors and Scholarships

- 2017 PhD Scholarship, Heinrich Böll Stiftung
- 2016 Dean's List, Hertie School of Governance

Funding

- 2024–2026 **DESTINY**, *Co-applicant and work package lead*, £10,000,000 - funded by the Wellcome Trust
- 2024–2026 **Pathfinder 2**, *Co-applicant*, £3,500,000 - funded by the Wellcome Trust
- 2023–2024 **Digital Evidence Synthesis Tools in Climate and Health**, *Co-applicant*, £320,000 - funded by the Wellcome Trust

2021–2022 **Climate Vulnerability Monitor**, *Design and lead of Work Package 1 - Observed climate change*, €669,390 - funded by Climate Vulnerable Forum

Teaching Experience

- Autumn 2023 **Methods in Planetary Health**, *Planetary Health MsC*, London School of Hygiene and Tropical Medicine, London, Guest Lecturer (1 Session)
- Autumn 2023 **Text as Data**, *Data Science for Public Policy MsC*, Hertie School of Governance, Berlin, Guest Lecturer (12 Sessions)
- Autumn 2022 **Text as Data**, *Data Science for Public Policy MsC*, Hertie School of Governance, Berlin, Guest Lecturer (12 Sessions)
- Summer 2022 **NLP for evidence synthesis in climate change research**, *Climate Change AI Summer School*, Online, Tutorial

Mentoring Experience

- 2021 **Co-supervision of Master Thesis**, *The dynamics of hate speech against Greta Thunberg: An analysis of Tweets*, Nina Locher, Hertie School of Governance

Research Service

- Peer Review I have provided reviews for Nature Climate Change, Nature Sustainability, Energy Research & Social Science, Environmental Research Letters, Climatic Change, Regional Environmental Change, Research Synthesis Methods, The Journal of Hydrology, and the Journal of the American Medical Informatic Association
- Assessments I was a contributing author in Chapter 9 of Working Group II's contribution to the sixth assessment report of the Intergovernmental Panel on Climate Change. I was a contributing author in Chapter 3 of the Austrian Panel on Climate Change's special report "Strukturen für ein klimafreundliches Leben"

Media Echo

My work has been covered by the Washington Post, CBC, CNN, Al Jazeera, ABC News, and the Frankfurter Allgemeine Zeitung.

Work in Progress

- Climate policy What is the evidence on climate mitigation policies, and to what extent can it be identified and classified using Machine Learning?
- Climate and Health Attributing the impacts of climate change on health to human influence on the climate
- ML-assisted review Biased urns for precise stopping criteria in machine learning-assisted systematic review screening

Peer Reviewed Publications

Up-to-date publication statistics are available at <https://scholar.google.com/citations?user=B6mNFrAAAAAJ&hl=en>).

- [1] Callaghan, M. et al. Machine learning map of climate policy literature reveals disparities between scientific attention, policy density, and emissions. *npj Climate Action*, 4(1):1–14, February 2025. ISSN 2731-9814. doi: 10.1038/s44168-024-00196-0.
- [2] Döbbling-Hildebrandt, N. et al. Systematic review and meta-analysis of ex-post evaluations on the effectiveness of carbon pricing. *Nature Communications*, 15(1):4147, May 2024. ISSN 2041-1723. doi: 10.1038/s41467-024-48512-w.
- [3] Callaghan, M. et al. Computer-assisted screening in systematic evidence synthesis requires robust and well-evaluated stopping criteria. *Systematic Reviews*, 13(1):284, November 2024. ISSN 2046-4053. doi: 10.1186/s13643-024-02699-7.
- [4] Young, S. et al. Collaboration and term usage dynamics in agricultural buffer strip research: Research weaving. *Ecological Solutions and Evidence*, 5(3):e12376, 2024. ISSN 2688-8319. doi: 10.1002/2688-8319.12376.
- [5] Danilenko, D. et al. Women are under-represented in adaptation policy research and are more likely to emphasise justice topics. *Environmental Research: Climate*, 3(4):045010, August 2024. ISSN 2752-5295. doi: 10.1088/2752-5295/ad6f3b.
- [6] van Daalen, K.R. et al. The 2024 Europe report of the Lancet Countdown on health and climate change: Unprecedented warming demands unprecedented action. *The Lancet Public Health*, 9(7):e495–e522, July 2024. ISSN 2468-2667. doi: 10.1016/S2468-2667(24)00055-0.
- [7] Cai, W. et al. The 2024 China report of the Lancet Countdown on health and climate change: Launching a new low-carbon, healthy journey. *The Lancet Public Health*, 9(12):e1070–e1088, December 2024. ISSN 2468-2667. doi: 10.1016/S2468-2667(24)00241-X.
- [8] Romanello, M. et al. The 2024 report of the Lancet Countdown on health and climate change: Facing record-breaking threats from delayed action. *The Lancet*, 0(0), October 2024. ISSN 0140-6736, 1474-547X. doi: 10.1016/S0140-6736(24)01822-1.
- [9] Repke, T. et al. How global crises compete for our attention: Insights from 13.5 million tweets on climate change during COVID-19. *Energy Research & Social Science*, 116:103668, October 2024. ISSN 2214-6296. doi: 10.1016/j.erss.2024.103668.
- [10] Müller-Hansen, F. et al. Attention, sentiments and emotions towards emerging climate technologies on Twitter. *Global Environmental Change*, 83:102765, December 2023. ISSN 0959-3780. doi: 10.1016/j.gloenvcha.2023.102765.
- [11] Zhang, S. et al. The 2023 China report of the Lancet Countdown on health and climate change: Taking stock for a thriving future. *The Lancet Public Health*, 8(12):e978–e995, December 2023. ISSN 2468-2667. doi: 10.1016/S2468-2667(23)00245-1.
- [12] Repke, T. et al. Attention to climate change only temporarily diverted by COVID-19. *Submitted to Environmental Research Communications*, 2023.
- [13] Romanello, M. et al. The 2023 report of the Lancet Countdown on health and climate change: The imperative for a health-centred response in a world facing irreversible harms. *The Lancet*, 402(10419):2346–2394, December 2023. ISSN 0140-6736, 1474-547X. doi: 10.1016/S0140-6736(23)01859-7.

- [14] Smith, S.M. et al. The State of Carbon Dioxide Removal - 1st Edition. Technical report, The State of Carbon Dioxide Removal, 2023.
- [15] Romanello, M. et al. The 2022 report of the Lancet Countdown on health and climate change: Health at the mercy of fossil fuels. *The Lancet*, 400(10363):1619–1654, November 2022. ISSN 0140-6736, 1474-547X. doi: 10.1016/S0140-6736(22)01540-9.
- [16] Müller-Hansen, F. et al. The German coal debate on Twitter: Reactions to a corporate policy process. *Energy Policy*, 169:113178, October 2022. ISSN 0301-4215. doi: 10.1016/j.enpol.2022.113178.
- [17] Callaghan, M. et al. What is the evidence on climate mitigation policies, and to what extent can it be identified and classified using Machine Learning? A machine-learning-assisted systematic map protocol. 2022. doi: 10.21203/rs.3.pex-1910/v1.
- [18] Creutzig, F. et al. Demand-side solutions to climate change mitigation consistent with high levels of well-being. *Nature Climate Change*, 12(1):36–46, January 2022. ISSN 1758-6798. doi: 10.1038/s41558-021-01219-y.
- [19] van Daalen, K.R. et al. The 2022 Europe report of the Lancet Countdown on health and climate change: Towards a climate resilient future. *The Lancet Public Health*, 7(11):e942–e965, November 2022. ISSN 2468-2667. doi: 10.1016/S2468-2667(22)00197-9.
- [20] Cai, W. et al. The 2022 China report of the Lancet Countdown on health and climate change: Leveraging climate actions for healthy ageing. *The Lancet Public Health*, 0(0), October 2022. ISSN 2468-2667. doi: 10.1016/S2468-2667(22)00224-9.
- [21] Fisch-Romito, V. et al. Systematic map of the literature on carbon lock-in induced by long-lived capital. *Environmental Research Letters*, April 2021. ISSN 1748-9326. doi: 10.1088/1748-9326/aba660.
- [22] Creutzig, F. et al. Reviewing the scope and thematic focus of 100,000 publications on energy consumption, services and social aspects of climate change: A big data approach to demand-side mitigation. *Environmental Research Letters*, 16(3):033001, February 2021. ISSN 1748-9326. doi: 10.1088/1748-9326/abd78b.
- [23] Berrang-Ford, L. et al. Systematic mapping of global research on climate and health: A machine learning review. *The Lancet Planetary Health*, 5(8):e514–e525, August 2021. ISSN 2542-5196. doi: 10.1016/S2542-5196(21)00179-0.
- [24] Diluiso, F. et al. Coal transitions—part 1: A systematic map and review of case study learnings from regional, national, and local coal phase-out experiences. *Environmental Research Letters*, 16(11):113003, October 2021. ISSN 1748-9326. doi: 10.1088/1748-9326/ac1b58.
- [25] Callaghan, M. et al. Machine-learning-based evidence and attribution mapping of 100,000 climate impact studies. *Nature Climate Change*, 11(11):966–972, November 2021. ISSN 1758-6798. doi: 10.1038/s41558-021-01168-6.
- [26] Berrang-Ford, L. et al. A systematic global stocktake of evidence on human adaptation to climate change. *Nature Climate Change*, 11(11):989–1000, November 2021. ISSN 1758-6798. doi: 10.1038/s41558-021-01170-y.

- [27] Sietsma, A.J. et al. Progress in climate change adaptation research. *Environmental Research Letters*, 16(5):054038, April 2021. ISSN 1748-9326. doi: 10.1088/1748-9326/abf7f3.
- [28] Khanna, T.M. et al. A multi-country meta-analysis on the role of behavioural change in reducing energy consumption and CO2 emissions in residential buildings. *Nature Energy*, 6(9):925–932, September 2021. ISSN 2058-7546. doi: 10.1038/s41560-021-00866-x.
- [29] Scheelbeek, P.F.D. et al. The effects on public health of climate change adaptation responses: A systematic review of evidence from low- and middle-income countries. *Environmental Research Letters*, 16(7):073001, July 2021. ISSN 1748-9326. doi: 10.1088/1748-9326/ac092c.
- [30] Müller-Hansen, F. et al. Who cares about coal? Analyzing 70 years of German parliamentary debates on coal with dynamic topic modeling. *Energy Research & Social Science*, 72:101869, February 2021. ISSN 2214-6296. doi: 10.1016/j.erss.2020.101869.
- [31] Callaghan, M.W., Minx, J.C. and Forster, P.M. A topography of climate change research. *Nature Climate Change*, 10(2):118–123, February 2020. ISSN 1758-6798. doi: 10.1038/s41558-019-0684-5.
- [32] Haddaway, N.R. et al. On the use of computer-assistance to facilitate systematic mapping. *Campbell Systematic Reviews*, 16(4):e1129, November 2020. ISSN 1891-1803. doi: 10.1002/cl2.1129.
- [33] Müller-Hansen, F., Callaghan, M.W. and Minx, J.C. Text as big data: Develop codes of practice for rigorous computational text analysis in energy social science. *Energy Research & Social Science*, 70:101691, December 2020. ISSN 2214-6296. doi: 10.1016/j.erss.2020.101691.
- [34] Callaghan, M. and Müller-Hansen, F. Statistical Stopping Criteria for Automated Screening in Systematic Reviews. *Systematic Reviews*, September 2020. doi: 10.21203/rs.2.18218/v2.
- [35] Ivanova, D. et al. Quantifying the potential for climate change mitigation of consumption options. *Environmental Research Letters*, 15(9):093001, August 2020. ISSN 1748-9326. doi: 10.1088/1748-9326/ab8589.
- [36] Lamb, W.F. et al. Learning about urban climate solutions from case studies. *Nature Climate Change*, 9(4):279–287, April 2019. ISSN 1758-6798. doi: 10.1038/s41558-019-0440-x.
- [37] Hilaire, J. et al. Negative emissions and international climate goals—learning from and about mitigation scenarios. *Climatic Change*, 157(2):189–219, November 2019. ISSN 1573-1480. doi: 10.1007/s10584-019-02516-4.
- [38] Minx, J.C. et al. Negative emissions—Part 1: Research landscape and synthesis. *Environmental Research Letters*, 13(6):063001, May 2018. ISSN 1748-9326. doi: 10.1088/1748-9326/aabf9b.
- [39] Lamb, W.F. et al. The literature landscape on 1.5[deg]C climate change and cities. *Current Opinion in Environmental Sustainability*, 30:26–34, February 2018. ISSN 1877-3435. doi: 10.1016/j.cosust.2018.02.008.
- [40] Nemet, G.F. et al. Negative emissions—Part 3: Innovation and upscaling. *Environmental Research Letters*, 13(6):063003, May 2018. ISSN 1748-9326. doi: 10.1088/1748-9326/aabff4.

- [41] Fuss, S. et al. Negative emissions—Part 2: Costs, potentials and side effects. *Environmental Research Letters*, 13(6):063002, May 2018. ISSN 1748-9326. doi: 10.1088/1748-9326/aabf9f.
- [42] Minx, J.C. et al. Learning about climate change solutions in the IPCC and beyond. *Environmental Science & Policy*, 77:252–259, November 2017. ISSN 1462-9011. doi: 10.1016/j.envsci.2017.05.014.
- [43] Minx, J.C. et al. Fast growing research on negative emissions. *Environmental Research Letters*, 12(3):035007, March 2017. ISSN 1748-9326. doi: 10.1088/1748-9326/aa5ee5.
- [44] von Stechow, C. et al. 2[deg]C and SDGs: United they stand, divided they fall? *Environmental Research Letters*, 11(3):034022, March 2016. ISSN 1748-9326. doi: 10.1088/1748-9326/11/3/034022.

Non peer reviewed publications

- July 2022 **Here are the most effective things you can do to fight climate change**, *Guest post for the Conversation*, link
- November 2021 **What 100,000 studies tell us about climate impacts around the world**, *Guest post for Carbon Brief*, link

Software

- NACSOS **A Django site for managing collections of documents, screening or coding them, and doing NLP tasks with them like topic modelling or classification**, <https://doi.org/10.5281/zenodo.4121526>
- Climate and Health Map **An interactive site built with Dash to present the results of a machine learning-assisted systematic map of climate and health literature**, <https://apsis.mcc-berlin.net/climate-health/>
- The Homero-manteion **A digitized version of the ancient method of divination using Homeric poetry**, <http://homeromanteion.com/>
- BuscarR **An R package to support safe early stopping during machine learning prioritised screening**, <https://mcallaghan.github.io/buscarR/>
- Buscarpy **A Python package to support safe early stopping during machine learning prioritised screening**, <https://buscarpy.readthedocs.io/>

Conferences and Talks

- Poster Presentation **The safe use of LLMs for automated screening**, *Global Evidence Summit*, Prague, September 11 2024
- Invited Keynote **Opportunities and challenges in using NLP to assist evidence synthesis**, *Text-as-data in Hydrology, Natural Hazards, and Climate Change Research*, Potsdam, March 1 2024
- Poster Presentation **BUSCAR: Biased Urn-based Stopping Criteria for technology Assisted Reviews**, *Cochrane Colloquium*, London, September 6 2023
- Invited Talk **Natural Language Processing for climate change texts at the science-policy interface**, *Climate Change AI Berlin Brandenburg*, Berlin, July 6 2022

- Invited Webinar **Machine Reading the Growing Climate Science and Adaptation Literature**, *Climate Change AI*, Online, May 13 2022
- Invited Webinar **A database of 100,000 studies on attributable climate impacts derived with machine learning: methodology and potential uses or follow-up analyses**, *PROCLIAS - Process-based models for climate impact attribution across sectors*, Online, May 09 2022
- Invited Talk **Statistical stopping criterial for automated screening in systematic reviews**, *Evidence Synthesis & Meta-Analysis in R*, Online Conference, January 18 2021
- Conference Presentation **Machine learning for systematically mapping the climate change literature**, *Applied Machine Learning Days*, EPFL, Lausanne, February 01 2020
- Conference Presentation **A Topography of Climate Change Research**, *EGU*, Vienna, April 09 2019
- Conference Presentation **A Topography of Climate Change Research**, *EASST*, Lancaster, July 26 2018